Cooperation and Sociality

How can network characteristics serve as guidance to the decision-making in social cooperation?

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Abstract

The emergence of human cooperation is listed one of the pressing scientific puzzles that remain to be disentangled. Since the seminal work by Nowak and May (1992), modeling efforts on the evolution of cooperation has shifted the attention to the structure of networks, as was recently summarized in Nowak (2006). Network not only determines the topology that confines whom actors would interact with (network neighbors); the structural properties of a node that an actor resides also provides useful information to an actor's decision-making of whether to cooperate with network neighbors, thereby making it possible for cooperation to emerge. I illustrate this point in the talk by showing a recently published evolutionary simulation model, followed by some empirical evidences derived from psychologists' experimental studies. Future research along this interesting direction will be outlined and discussed.

Keyword: Evolutionary Game, Complex Network, Cooperation, Sociality

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References

[1] M. Nowak, and R. May, "Evolutionary games and spatial chaos", Nature vol.359, 826-829 (1992).

[2] M. Nowak, "Five Rukes for the evolution of Cooperation" Science vol.314, 1560-1563 (2006).